

Architectural Particular Specification

BUILDING MAINTENANCE UNIT

SECTION 11.3

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SECTION 11.3

BUILDING MAINTENANCE UNIT

PART 1 GENERAL

1.01 SUMMARY OF WORK

- A. The Contractor for the Building Management Unit (hereafter abbreviated BMU) shall include the design, supply, installation, testing, and commissioning and maintenance, of a Building Management Unit system for cleaning and maintenance of the following areas:-
 - 1. Facades and all external wall of tower portion
 - 2. Roof floors, including all inaccessible flat roofs
- B. The Works described in this Specification are to be provided by the Contractor. All Work shall comply with the requirements of the Contract Drawings and the Specification, and with all appropriate standards, Relevant Authorities' regulations and codes.
- C. The Contractor shall be responsible for the design, shop drawings, manufacture, supply, delivery, installation, testing, commissioning, as built drawings, maintenance manuals and warranties.
- D. The Works shall also include the training of operatives, provision of maintenance and operating manuals and provision of routine maintenance of the equipment for a period of the defined Defects Liability Period (DLP), commencing after the issue of the Completion Certificate under the Contract.
- E. The Contractor shall take full responsibility for the performance of the proposed system, regardless of the criteria stated in these documents.
- F. The Contractor is required to carry out the design of BMU and prepare all design documents for the Architect and Structural Engineer to review without objection. The design documents shall clearly delineate the nature and scope of the submittal, including the method and total equipment required to clean and access all exterior surfaces of the Tower as well as deliver materials and glazing where required.
- G. The Contractor shall be responsible for the maintenance of the proprietary Building Maintenance Equipment system(s) during the Defects Liability Period after the issue of the Completion Certificate under the Contract.
- H. Works shall include but not be limited to the systems itemized below. The locations of all equipment are generally identified on the Contract Drawings and in this Specification.
 - 1. Track mounted davit trolley system
 - 2. Wall mounted davit arms
 - 3. Monorail track system with powered cradle
- I. The Contractor shall be responsible for furnishing all anchors and inserts for installation during construction of any superstructure, or any anchors required in the façade during fabrication and construction of the façade. The Contractor shall be responsible for inspection and verification of layouts before concreting or fixing. Contractor shall also provide fixing lug or restraint sockets at the external wall.

- J. Final dimensions and quantities shall to be checked based on all the latest related approved drawings and construction drawings. Quantities shall be confirmed by the Contractor.
- K. The Contractor shall bear all expenses for testing, and obtaining all required approvals, permits and certificates from relevant authorities.
- L. Relevant Authority Submissions: The Contractor shall be responsible for providing any drawings, information, test results and calculations necessary to obtain all Relevant Authorities' approvals. This includes conducting all tests required by Relevant Authorities. The Contractor shall allow for providing all design or other documentation required for achieving Relevant Authority approvals for both the BMU equipment and for wider ranging building planning and other approvals. The Contractor shall allow for all time required in the coordination of achieving such approvals, and for working with and coordination with the Architect in achieving such Relevant Authority approvals.
- M. The Contractor shall provide all necessary temporary works, such as scaffolding, working platforms for the execution of the Works.
- N. The BMU must be capable of maintaining and servicing the largest or heaviest façade panel.
- O. The Works shall be fully tested, commissioned and handed over in running order and shall include not only the major items of plant and equipment shown or specified but all the incidental sundry components necessary for the proper operation of the installation, with their layout charges, whether or not these sundry components are mentioned in the Contract Document.
- P. Design Responsibility:
 - 1. The Specification and the Employer's Drawings define the scope, performance criteria, and strategy for the Building Maintenance Unit. All costs associated with the design, certification, approvals, checking and testing shall be the responsibility of the Contractor.
 - 2. Notwithstanding that drawings and documents are provided by the Employer, it is specifically noted that these are for reference only and indicate a possible method of achieving the required results. The Employer's Drawings are provided for information only. The Contractor shall take full responsibility for the performance of the proposed system, regardless of the criteria stated in these documents. This responsibility shall include coordination with designed features of the development to ensure that the proposed systems are fully functional and that full access is provided to all required areas, including of delivery of materials where required.
 - 3. The Contractor shall be solely responsible for the design of all the Building Maintenance Equipment systems, including relevant support structures, to be supplied under this Contract. The Contractor shall appoint a BMU designer to perform the engineering analysis who is a Registered Professional Engineer (RPE) under Cap. 409 Engineers Registration Ordinance with a minimum of 10 years of proven design and installation experience of BMU in a similar scale, using the performance requirements and design criteria indicated in this Specification and all in accordance with the standards, regulations and Relevant Authority requirements. This shall also include full calculations, technical information, and testing data in order to justify the provided works are of the quality required. All design calculations and drawings shall be further reviewed and certified by an Independent Checking Engineer (ICE) who is a Registered Structural Engineer (RSE) under Cap. 123 Buildings Ordinance as stipulated in the Particular Specification.

4. The Contractor shall coordinate with the Architect for the design of the building structure which supports the Access System and submit all relevant structural loading requirements of the Access System with the ICE certification.
5. The Building Maintenance Equipment shall be inspected, tested and certified by an Independent Third Party Authority in the factory prior to shipment to Hong Kong, as well as by an Independent Professional Engineer, who shall be a RPE in accordance with Cap. 409 Engineers Registration Ordinance, on-site prior to use and hand over of the equipment to the Employer. For clarity, the Independent Third Party Authority in the factory shall be an approved accreditation agency such as the TÜV or equivalent.
6. The Contractor shall be responsible for all submissions to Relevant Authorities and for obtaining all necessary certification to meet and satisfy statutory requirements.
7. The Contractor shall be responsible for coordinating all points of interface of the Works with other packages.
8. The Contractor shall incorporate emergency egress provisions in order to ensure maintenance personnel can safely depart from the cleaning or working location should the Building Maintenance Equipment malfunction.
9. The Access Systems shall provide access all parts of the tower and building structures below the highest element of the structures, including roof areas, recesses, soffits, ceilings and architectural features, for maintenance, façade panel / glass replacement and cleaning works.

1.02 QUALITY ASSURANCE

- A. All equipment shall be sourced, designed and manufactured from an internationally recognised supplier/manufacturer with quality control accreditation to ISO9001 standard or equivalent.
- B. The Works shall be executed by a qualified firm specializing in the design, fabrication and installation of Systems similar in quality and scope to that required on this project and having a minimum of 5 years documented production and installation experience.
- C. The manufacturer shall be able to show employment of experienced personnel, physical facilities, established quality control procedures, management capability sufficient to produce the required systems without causing Project delay.
- D. The manufacturer shall submit a list of at least three representative reference systems three years old or older, with date of installation and engineers' and Building Employers' names, addresses and phone numbers.
- E. Installer Qualifications: An experienced installer shall be engaged who has a minimum of five years' experience in installation of systems in Hong Kong similar to this project and who is an authorized representative of the equipment manufacturer for both installation and maintenance of the type of units required for this Project, and whose installations have resulted in construction with a record of successful in-service performance.
- F. Welder: Submit for the Architect and Engineer's information, evidence in compliance with AWS D1.1 that each welder has satisfactorily passed qualification tests for the welding processes involved and has undergone recertification if the welder has not welded for more than six months.
- G. Instruction and training in the operation and maintenance of the façade access equipment shall be provided to the Employer's representatives.

- H. Where requirements of governing codes, regulations, laws, and rules required by authorities having jurisdiction over this project conflict with these Specifications and are mandatory, they shall be followed the same as if fully specified herein.
- I. All equipment shall be verified that it has been installed correctly and that it functions properly; adjustments shall be made as necessary to ensure satisfactory operation.
- J. Full live load and operational tests shall be conducted after completion of the installation. This shall be under maximum design live loading conditions over a selected range of the building surfaces, in accordance with applicable standards.
- K. The manufacturer shall make available to the user the following information on one or more durable signs or plates mounted on the roof car and cradle.
 - 1. Name and address of the manufacturer
 - 2. Designation of series or type Serial number (if existing)
 - 3. Year of construction
 - 4. Rated load of the cradle and the maximum number of operatives
- L. At a time mutually agreeable to all parties, at least 2 full days shall be allowed to conduct operational demonstrations of each system for the Employer or Employer's representative, or such time as required until the Employer has a full understanding of the system. This shall be after completion of the commissioning and operational tests.
- M. Any components and deficiencies observed as a result of these tests and demonstrations shall be repaired, replaced or corrected and retested to assure compliance with the requirements of this Contract, relevant Standards such as EN1808, Cap. 59A Factories and Industrial Undertakings Regulations, and other relevant Standards, Codes and Regulations.

1.03 PERFORMANCE REQUIREMENTS

- A. The BMU shall be installed to carry two workmen inside each cradle including necessary equipment and material for complete cleaning and maintenance of window wall, aluminum window, and curtain wall at abovementioned areas;
- B. The dimension of the BMU shall be designed such that the workmen inside the BMU cradle can work reasonably safely and comfortably for cleaning every external part of the curtain walls, window walls, cladding, architectural features, and signage as shown on the elevation drawing and plans of the proposed building and also enable the workmen to take off and fixing of the glass, mullion, transom, and metal panel for replacement from outside of the building;
- C. The BMU shall be designed for a nominal vertical travelling speed of not exceed 10m/min;
- D. The cradle shall be balance disregarding the load imposed by the workmen & their cleaning equipment;
- E. The proposal should ensure that enough allowance has been allowed for avoiding conflict between the vertical movement of the cradle and the protruding sunshading devices, building services and architectural features along typical external façade;
- F. Provide water and electrical supply from roof floor to cradle for cleaning and maintenance purpose.
- G. All works and materials shall comply with all latest requirements, regulations, recommendations and codes of practice including but not limited to the followings:-

1. Code of Practice on Access for External Maintenance 2021
 2. Code of Practice for safe use and operation of suspended working platforms
 3. The 15th Edition of IEE Wiring Regulation
 4. EN1808 : 1999 safety requirements of suspended access equipment
 5. Requirement for operation the gondola system by Labor Department, Electrical and Mechanical Services Department of Hong Kong Government
 6. Construction Sites (Safety) Regulations, 1987
 7. British Standard BS6037 : 2004 – Code of Practice for Permanently Installed Suspended Access Equipment
 8. Building Authority – building regulations
 9. Fire Service Department – Codes of Practice and current requirements including latest FSD circulars
 10. Civil Services Department
 11. Environmental Protection Bureau
 12. The Wiron3 Regulations of the Institute of Electrical Engineers, Hong Kong
 13. Any other authorities having jurisdiction over the installation.
- H. The Contractor shall comply as a minimum with all current regulations, codes of practice and references relevant to Building Maintenance Equipment systems in Hong Kong for the BMU. Where there are no relevant Standards or codes of practice in Hong Kong, then International Standards shall be applied. All work and labour practices under this contract shall be in strict accordance with all applicable building Codes, Regulations and Standards prevailing in Hong Kong. It is the responsibility of the Contractor to use and comply with the latest version of any standard or regulation that may be applicable. The list below is for guidance purposes only and is a minimum requirement.
- I. Test certificate shall be obtained from the Chartered Surveyor in Hong Kong to certify the BMU is in good operating condition prior to handover
- J. All expenses of test, permits, stamping of samples and obtaining certificates shall be responsible by the contractor.
- K. The installation materials and equipment shall comply with the latest requirement of the Standard Codes, Guides and other documents issued by the Authorities, Institutions and Organizations referred to in various sections of the specifications, including the following:-
- | | |
|------|---|
| BSI | British Standards Institute |
| CIBS | Chartered Institute of Building Services |
| AISI | American Iron & Steel Institute |
| FOC | Fire Offices Committee |
| IEE | Institute of Electrical Engineers |
| ISO | International Standards Organization |
| ASIM | American Society of Testing and Material |
| NBS | National Bureau of Standards |
| NEMA | National Electrical Manufacturers Association |
| NFPA | National Fire Protection Association |
| DIN | International Standard |

IEC International Electrotechnical Committee

NEC National Electrical Code

In case of discrepancies between requirements of the above and local codes and legislations. The local codes and legislations shall be followed.

L. Builder's Work Drawings

The Contractor is to ensure all holes, chases, etc. to be left in the structural building works or any external works as shall be left in as works proceed. No subsequent cut-out at the structural building works or any external works shall be allowed, except in so far as may be necessary due to subsequent authorized variations to the scheme.

M. The Access System equipment shall be designed to withstand 30 km/h wind velocities while being used for normal operations, including for any uplift. They shall be able to withstand 240 km/h wind velocity, or that indicated by the wind pressure study whichever is more stringent, when in their secured stored positions. Hold down "Typhoon Anchors" shall be provided at the parking locations for all Building Maintenance Equipment / BMUs and other associated equipment.

N. All anchor components shall conform to proper engineering principles and shall be designed and engineered by a BMU designer as stipulated in this specification, qualified in the design of Building Maintenance Equipment and its application, and shall comply with all safety requirements.

O. Structural steel supports and all related work necessary to provide a complete installation for each Building Maintenance Equipment System shall be designed and engineered in accordance with good engineering practice and with relevant Standards in order to provide a complete installation. The total system shall be compatible with the building structure and the Structural Engineer shall be notified if incompatibility exists. Imparted loads to load carrying members of the building from the attachment shall be within allowable limits. No member shall be overstressed. Loads imposed on the structure shall be submitted to the Structural Engineer for review without objection and coordination with Other Contractors and disciplines.

P. Assemblies shall allow for thermal movements resulting from the maximum change (range) in ambient and surface temperatures. Engineering calculations shall be based on surface temperatures of materials due to both solar heat gain and night time heat lost.

Q. All apparatus and equipment shall be so designed and constructed that they operate satisfactorily and without any deleterious effect for prolonged and continuous periods in the applicable climatic conditions and ambient temperature conditions.

1.04 ENVIRONMENTAL REQUIREMENTS

(NOT USED)

1.05 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITIONS

A. The Contractor shall take account of the climatic conditions prevailing in Hong Kong and to be aware of the high temperature and the high relative humidity during the summer and the contracting low temperature and low relative humidity experienced in the winter.

B. The Contractor in proposing equipment for the installation shall warrant that all materials and items of equipment are suitable for continuous use and/or operation in the various climatic conditions encountered.

C. All the electrical, electronic and mechanical equipment of the installation shall be at least suitable for use in service conditions as stated below:

1. Climate Tropical
2. Temperature range 0 degree Celsius to 55 degree Celsius
3. Relative humidity 100% maximum

1.06 SUBMITTALS

- A. The Contractor shall submit a design proposal together with detail description of the BMU including performance and operational procedure to the Architect for approval prior to installation, fabrication and ordering of equipment. The design proposal shall be in full compliance with the statutory requirements of the Building Authority. Preparation and submission of information to the Building Authority as may be required shall be included in this contract. Such information are to be endorsed by the project RSE subject to his satisfaction for BD submission.
- B. Submission requirement
 1. The Contractor shall, as soon as possible and well before the relevant work proceeds, prepare and submit four (4 nos.) sets for approval co-ordinated working drawings showing sequence of work. Equipment fabrication (shop), detailed internal and external wiring diagrams, etc. The drawings are to be based on tender drawings, but are to be amended to account for any modifications to the building or installation which may have taken place, and for any adjustments due to the actual equipment offered. The working drawings shall be to suitable scales relative to their purpose and are to include adequate plans, elevations, sections and view as necessary to show all dimensions, levels and installation details.
 2. The contractor shall also submit the followings for Architect and Engineers' review and approval:
 - a. Structural calculations and detailed structural requirements for the building structure, including operation weights and loading of all parts of the proposed systems.
 - b. Detailed electrical requirements, both temporary and permanent, for all parts of the proposed systems.
 - c. Detailed lightning protection earth bonding lock mechanism and earth monitoring system.
 - d. Detailed schedule of material and equipment.
 - e. A proposed list of all tools and spares.
 - a. Detailed Inspection and Testing Plan for both factory and site to be carried out in accordance with the requirements of the Standards and the Specification.
 3. The contractor shall be responsible for verifying the accuracy of all dimensions abstracted from Tender drawings or Architect drawings and used in the preparation of his drawings.
 4. Submitted drawings shall be based on Employer's Drawings but amended to take account of any modifications to the tower or installation which may have taken place, and of any adjustments due to the actual equipment selected. The working drawings shall be to suitable scales relative to their purposes, and shall include adequate plans, elevations, sections and views as necessary to show all dimensions, levels and installation details. All such drawings shall be reviewed without objection by the Architect before any work is put in hand.
 5. The Contractor shall provide fully detailed builder's work drawings. These drawings shall show dead and live loads of all plant/equipment and fully dimensioned details of all plant/equipment bases, wall chases, and penetrations. This works shall be incorporated in the Works programme.

6. The Architect may disapprove, approve, or amend such drawings. No claim will be accepted for disapprovals or amendments required by the Architect. The approval by the Architect of any such drawings shall not relieve the Contractor of his duties and responsibilities under contract. The contractor shall notify the architect to inspect the routing of the BMU or mock-up in the beginning of works prior to the execution of the works. In failure to comply with advance notification, the Architect is entitled to alter the work at the cost of the Contractor notwithstanding of the approval of shop drawings.
7. The contractor shall allow sufficient time for approval of shop drawings and samples. No claim for time will be entertained due to Contractor's failure to submit shop drawings or materials timely for approval.
2. The contractor shall prepare as-built plans, operation and maintenance instructions and submit to the Architect. Four (4 nos.) sets or greater number as required elsewhere in contract are to be provided.

C. Dimensions

Figured dimensions are to be taken. Do not scale or measure from drawings. Before commencing any work or ordering any materials the Contractor must verify all measurement. If any discrepancies are found they must be clarified with the Architect immediately.

D. Operation procedure and training

1. Operation Procedure

The Contractor shall include in his design proposal an operation procedure and method statement describing procedure and method of using each BMU system for the maintenance from the roof and the whole operation from start to finish including maintenance instructions for the use of the proposed BMU. Three (3) nos. of maintenance and operation manuals are required to be submitted to the Architect.

2. Operation and Maintenance Manual

The operation/maintenance manuals shall contain the following information:

- a. Basic information including but not limited to the name of development, title of document, name and address of the supplier, contact telephone numbers for normal and emergency use etc.
- b. Description of the installation
- c. Schedule details of all equipment and plant.
- d. Operating instructions including details of any automatic control system.
- e. All contents of instruction shall comply as a minimum with EN1808, Hong Kong Factories and Industrial Undertakings Regulations for BMUs, and other relevant Standards, Codes and Regulations.
- f. Maintenance and fault-finding instructions including composite schedule for routine maintenance. The maintenance schedule shall include details such as frequency of servicing, overhauling, replacement, etc.
- g. Test reports and commissioning records.
- h. Recommended spare parts and lubricants.
- i. List of equipment and plant with manufacturer's name and address and local agent, if applicable.
- j. Photographs of all key equipment and controls, suitably annotated for easy understanding.
- k. Manufacturer's literature suitably indexed to include shop drawings, wiring diagrams, performance curves, etc.

- l. As fitted final wiring diagrams.
 - m. List of as-fitted drawings.
 - n. Glazing / claddings replacement methodology
 - o. All other necessary information for proper maintenance and operation.
 - p. As-fitted drawings showing the complete installation including all necessary details for the proper maintenance and operation of the system.
3. Tools
- a. The Contractor shall provide a complete set of recommended and required tools within a lockable tool box for maintenance and service of equipment within this Contract.
 - b. All tools and maintenance equipment shall be provided in original packing, clearly labelled and referenced.
 - c. All tools and maintenance equipment shall be as recommended by the equipment manufacturer and shall be subject to the final approval of the Architect.
4. Training
- The Contractor shall provide adequate training and at no less than 3 sessions for the Employer's staff until they are fully familiar with the operation and maintenance of the complete installation. The training shall include class room lectures supported by training manuals and audio-visual aids, and field demonstrations. The Contractor shall ensure that his training scheme(s) are suitable for the management and engineering staff of the Employer. The training course shall include all aspects of daily operation, plant limitations, emergency operation and repair. The content and formats of the training sessions shall be agreed with the Architect prior to actual session.

1.07 PRE-INSTALLATION CONFERENCE

A. Coordination

1. The Contractor shall be responsible for co-ordination with the Architect to provide a complete installation works in accordance with the requirements of this Specification. In particular, the works of the Contract shall interface with the following:
 - a. Concrete and Steel Structure:

The Contractor shall design, co-ordinate supply, survey, set out and install all structural support systems, holding down bolts, cast-in items and concrete/steel anchor systems and any other fixtures for the Building Maintenance Equipment. Then location of track plinths shall be designed by the Contractor and submitted to the Architect for review without objection.
 - b. Structural Framing:

Structural steel and concrete framing for buildings, facades and façade features, including any structural support systems for the Building Maintenance Equipment shall be by the Contractor. The design, co-ordination, supply, surveying, setting out and installation of any plinths or track fixing points, fixing bolts, anchor systems and other fixtures for the Building Maintenance Equipment shall be performed by the Contractor.
 - c. Equipment Loads:

The assumed equipment load is shown on the Employer's Drawing. Early definition and coordination by the Contractor is required to ensure that the loads are correctly defined and provided to the Architect for review without objection

- d. Façade coordination:
The design of BMU systems shall be coordinated with the design of the façade.
 - e. All hose retainers to the Building Maintenance Equipment shall be provided by the Contractor. The Contractor shall coordinate with the Architect regarding locations and the operational requirements.
 - f. Any cast in anchors shall be designed, manufactured and supplied for installation in the structural members. The location of the cast in anchors shall be defined by the Contractor.
 - g. Concrete or steel plinths/brackets, if required, shall be constructed by the Contractor. The locations of the plinths shall be defined by the Contractor and coordinated and approved by the Architect. The Contractor shall be responsible for inspection and verification of layouts and locations before concreting. The Contractor shall be responsible for grouting of plinths, when required, after installation and final levelling.
 - h. Glazing and façade panel replacement delivery strategy: The replacement delivery strategy shall be developed by the Contractor to suit the requirements of the façade components.
 - i. Delivery of Equipment: The Contractor shall incorporate this in the works programme
- B. The Contractor shall ensure that staging of works and thorough interfacing coordination with the other packages are carried out during the design, fabrication, installation and commissioning of the Works.

1.08 DELIVERY STORAGE AND HANDLING

- A. Equipment and materials shall be delivered in the manufacturer's original, unopened, undamaged containers with identification labels intact.
- B. Equipment and materials shall be stored protected from exposure to harmful weather conditions and at temperature and humidity conditions as recommended by the material manufacturer.
- C. The Contractor shall be responsible for delivering all equipment into the assigned locations and incorporate this in the Works programme. The Contractor shall liaise with others to ensure the proper delivery route and programme.
- D. The Contractor shall be responsible for any costs to the client for any visits to the manufacturing premises to check on the progress of the manufacture of the BMUs, arising out of continuous delays on the part of the Contractor to provide verifiable manufacturing and construction details.
- E. The Contractor shall be fully aware of the space and structure constraints before actual delivery of equipment to site.
- F. The Contractor shall provide drawings showing in detail the proposed hoisting method prior to the hoisting operation together with a detailed method statement.
- G. The Contractor shall notify the Architect at least 30 days prior to the date that the equipment is due to be delivered to site. The expected delivery date and the programme for the installation of the machines and associated equipment shall be incorporated into the Works programme.

1.09 MOCK-UPS

(NOT USED)

1.10 TESTING

A. General

1. The Contractor shall carry out test to the equipment and plant installed to ensure that the operating directions and control sequences are correct. Loading test to the cradle gondola cage shall be carried out upon completion by the Contractor to ensure minimum loading requirement is achieved including tests for the adequacy and the efficient functioning of control and safety devices. All controls shall be tested in conjunction with all equipment concerned to ensure a normal and satisfactory operation of the whole plant.
2. The Building Maintenance Equipment shall be inspected, tested and certified by an Independent Third Party Authority in the factory prior to shipment to Hong Kong, as well as by an Independent Professional Engineer, who shall be a RPE in accordance with Cap. 409 Engineers Registration Ordinance, on-site prior to use and hand over of the equipment to the Employer. For clarity, the Independent Third Party Authority in the factory shall be an approved accreditation agency such as the TÜV or equivalent.
3. The Contractor shall submit test certificates issued by the manufacturer for the equipment to be used for this project (including type test certificate). All tests shall be certified by an Independent Professional Engineer experienced in the testing of Building Maintenance Equipment in accordance with the standards to which the system has been designed and manufactured, as well as any requirements of Hong Kong regulations or codes.
4. On completion of the works, the Contractor shall commission the Systems installed and put them into operation as fully operational Fit-For-Purpose Systems.
5. The Contractor shall be responsible for any costs of witnessing of testing and commissioning of equipment in both the factory and onsite by the Architect.
6. The Contractor shall obtain all necessary certificate/permit from relevant Authority for operation. All costs for testing and obtaining all permits and certificates from relevant authorities required to satisfy Relevant Authorities and to the requirements of the Standards specified herein shall be included in the Contract.
7. During commissioning and prior to the issue of the Completion Certificate, the Contractor shall provide all necessary facilities and instruments to carry out all tests as required to satisfy Relevant Authorities and the Architect that the installation meets the requirements of the Specification and Standards.
8. The Contractor shall submit test certificates by the Independent Professional Engineer certifying that the Building Maintenance Equipment system is in good and safe operating condition prior to the handover of the system. The name of the Independent Professional Engineer shall be submitted for approval by the Architect three months prior to the date of the testing.
9. The Contractor shall provide minimum 12 months or longer duration as stipulated elsewhere in contract. The free maintenance period shall commence from the date of the Certificate of Substantial Completion or the date of BMU handover, whichever is later.
10. During the Defects Liability Period, the Contractor shall carry out a regular comprehensive maintenance service including but not limited to systematic examination, lubrication and repair or replace of the parts wherever required. This shall be at no additional cost to the Employer.
11. In addition, the Contractor shall carry out all adjustments necessary for the safe, reliable and satisfactory operation of all the equipment supplied under the Contract.

B. Site Tests

1. The Architect shall have power to inspect all work in progress and upon completion and to require the Contractor to carry out tests in the presence of his authorized representative in order to prove that all work carried out and all material, apparatus and equipment installed are wholly satisfactory and fully meet the requirements of the Contract Document. All costs and charges associated with or arising from such inspections and tests, including costs incurred by the Contractor in carrying out the prescribed tests and in attending with the Architect whilst tests and inspections are in progress shall be borne in full by the Contractor. Any costs associated with further testing as the result of abortive tests or non-compliance of tests with the required standards shall be the responsibility of the Contractor.
2. The Contractor shall provide a comprehensive list of tests to be undertaken for review at least 28 days prior to the proposed date for the testing of the BMU. Additional costs for the Architect's representative to re-inspect any tests due to abortive tests or noncompliance of tests with the required standards shall be the responsibility of the Contractor.
3. Subject to the Architect's approval, third party testing shall be carried out on site by the Independent Professional Engineer.
4. Tests not carried out in the presence of the Architect or his authorized representative shall not be regarded as valid for the purpose of the Contract, unless the Architect shall have previously signified his inability to attend and shall have authorized the Contractor to proceed in his absence.
5. The Contractor shall ensure that the Architect or his authorized representative is provided with adequate facilities for carrying out such inspections as he may consider necessary, and that all tests are carried out in a safe and workmanlike manner complying with all relevant statutory safety requirements.
6. No material, apparatus, equipment, or installation shall be covered or otherwise permanently concealed from view until the Architect has had the opportunity to inspect it or has either has formally waived his right to inspect it.
7. No inspection or testing by the Architect or his authorized representative, nor the witnessing of satisfactory testing shall in any way relieve the Contractor, nor shall it in any way limit the right of the Architect to reject such items if they subsequently prove to be defective or unsatisfactory or unsuitable for their intended purpose.
8. The Contractor shall be wholly responsible for time scheduling his work so that inspection and tests can be carried out as specified and the prescribed notice given to the Architect without delaying the completion of the Works beyond the date set out in the approved construction programme. No apparatus, equipment, plant, or installations will be recognized as complete and ready for taking over until after all of the specified inspection and tests have been satisfactorily carried out.
9. The Architect shall have the right to waive, at his sole discretion, the requirements in relation to inspection of materials, apparatus equipment and installations and the witnessing of tests thereon. However, the exercise of this right by the Architect shall not in any way relieve the Contractor of his obligations to carry out the prescribed tests and to submit test certificates.
10. If any fault or defect should come to light during the Defects Liability Period in any of the apparatus, plant, equipment or installations, the Architect shall have the right to require the Contractor to carry out further tests to isolate the fault or defect and subsequently to prove that the fault or defect has been remedied. The cost of such testing shall be borne by the Contractor, including the cost of further inspections by the Architect.

C. Test Certificates and Commissioning Records

1. The Contractor shall submit three copies of test certificates to the Architect bearing the signatures of the Independent Professional Engineer. These are to

be in accordance with the requirements of Hong Kong Regulations and with prevailing standards.

2. Test certificates shall serve as a certified record that the item referred to has been shown under test to meet the requirements of the Contract.
3. The Independent Professional Engineer shall inspect and certify in writing the completion and safety of the Building Maintenance Equipment system installation and operation as specified. This shall be upon completion of the installation, prior to issue of Completion Certificate.
4. The Building Maintenance Equipment systems shall be tested, commissioned, and witnessed to the satisfaction of the Architect regarding the specified performance. The Contractor shall submit a testing and commissioning schedule two months before the actual testing and commissioning works for the Architect's review without objection.
5. Project record ("as-built") drawings showing actual installed locations and configuration shall be submitted. All changes to original design criteria and other specification requirements shall be recorded and documented. Wiring diagrams showing all electrical connections of equipment, including a legend sheet, shall be included in the "as built" drawings.

1.11 WARRANTY

This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.

1.12 SPARE PARTS

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

The Contractor shall provide a table of spare parts, consumable parts and special tools. The Contractor shall maintain a minimum amount of spares to meet the requirements of one year's operation.

The Employer shall reserve the right to purchase any additional quantity of spare and consumable parts in accordance with the quoted rates. The information supplied in respect of each spare part, consumable parts or special tools shall include but not be limited to the following:

1. The manufacturer's part number,
2. Space for the Employer's part number,
3. A full description of the spare part with an illustrated Part List,
4. Quantities to be supplied,
5. Expected utilization in twelve months,
6. Overall dimensions including packing (if any) for shelf space purposes,
7. A note as to inter-changeability or otherwise with similar parts,
8. A unit CIF to the site,
9. The source of the spare part or special tool, giving the manufacturer's name and address together with that of his Hong Kong agent,
10. The normal manufacturing and shipment lead times for additional quantities.

PART 2 PRODUCT

2.01 APPROVED MANUFACTURER

Subject to compliance with the Project Contract Documents, provide panel ceilings and required accessories from one or more of the following manufacturers, or equivalent approved by the Architect:

- A. China King Engineering Ltd.
Tel: (852) 2797 8699 / (852) 9738 3803
jasonlo@chinaking.com.hk
- B. Arrowtech Engineering (HK) Co. Ltd
Tel: (852) 2364 0727 / (852) 9059 4649
alex.to@arrowtech.com.hk
- C. Or equivalent approved by Architect

2.02 MATERIALS / SYSTEMS

- A. General
 - 1. Products and/or systems specified in this section establish the minimum functional, aesthetic, and quality standards required for work of this section.
 - 2. Works shall be executed by a manufacturer specializing in the design, fabrication, and installation of the Systems detailed in this Specification and who have demonstrable documented experience in providing such Systems for structures of this complexity and height.
 - a. System A: Track mounted Davit Trolley System
 - b. System B: Wall-mounted Davit Arm
 - c. System C: Monorail Track System with Powered Cradle

- B. System A - Track mounted Davit Trolley System

The system traverse on the GMS tracks mounted on parapet walls. The base frame has four wheel sets for the trolley to the two front wheel assemblies are centrically mounted.

- 1. Design Constraints
 - a. The proposed BMU systems shall be located on the building in conformance with tender drawings.;
 - b. The Contractor shall take full consideration of the presence of E/M equipment along with architectural features in designing the BMU system of the location, size and details. The architectural features and/or equipment are shown on contract drawings.
 - c. The location and siting of the BMU system shall not reduce to the architectural features and/or equipment under any circumstance;
 - d. The tenderer shall take full consideration of the height of the parapet wall, architectural features and equipment in relation to the level at which the BMU is to be mounted.

e. Technical data:-

- 1) Dimensions, locations, positions, system type and the like shall be proposed by the Tenderer with structural calculations to be approved by the Architect;
- 2) The whole gondola unit when in a parked position shall be capable of withstanding the wind load when typhoon signal No.10 is hoisted. BMU Operating wind speed shall not exceed 31m/s;
- 3) All movements of the BMU shall be fully protected by failsafe limit switches of protection class IP65. Key switches, emergency stop buttons and a lockable isolator are required;
- 4) All electrical and mechanical equipment incorporated in the BMU system, including driving motor, cables, control devices and the like shall be made to and comply with relevant Standards. All motors shall be European brand and comply with BEC-2018;
- 5) Other technical data not stated in the above but which the Tenderer considers relevant shall be clearly stated.
- 6) Any electrical and mechanical systems proposal to be used, items, equipment, accessories and the like shall be detailed in full and tabled for the Architect and M&E engineer's consideration and checking;

2. Particular Specifications

- a. The equipment is required at the locations shown on the Drawings to provide safe access for maintenance. All equipment is to be fully designed, engineered and detailed in accordance with the approved standards and codes. A full layout of the BMU and its operating locations with loadings is required for review.
- b. Access to the locations of the BMU shall be coordinated with the Employer, Main Contractor or Architect. Final dimensions and quantities shall be checked based on all the latest related construction drawings. Quantities shall be confirmed by the BMU Sub-Contractor;.

General Requirements	
Mounting Location	Parapet Mounted at R/F
Permanent	Yes
Quantity	Refer to tender drawings
Warranty Period	10 Year
Design Life (Fatigue etc.)	10 Years
Defects Liability Period	12 Months
BMU	
Operation	with 2.0m self-climbing type cradle attached
Height from top of roof	Proposed by tenderer
BMU Type	Davit Trolley System
Jibhead Type	Swiveling
Traverser	Run on Twin Parapet Rails type
Lifting Height	Refer to tender drawing
Machine Housing Slewing	Required
Material	Galvanized mild steel painted to architect's requirement
Suspended Load	250kg S.W.L. for gondola cradle
Total Jib Reach	Reaching the required extend
Height of Suspension Point	Proposed by tenderer
Outreach	1.8m nominal
BMU parking dimensions	Proposed by tenderer
Cradle	
Length	2.0m nominal
Width	0.7m nominal

Safe Working Load	250kg for personnel
Quantity	1 for BMU operation
Material	Aluminum alloy
Hoist speed	The hoisting and lowering speed shall not exceed 10m/min
Number of persons	2
Track	
Gauge	Proposed by tenderer
Track Type	Proposed by tenderer
Material	Galvanized mild steel
Length	Refer to tender drawing
Rail Traverser Type	Parapet mounted
Power Supply	
Supply	380V 3 Phase, 32A, 50Hz
Sockets	PCE 5 pin 32 Amp

C. System B: Wall-mounted Davit Arm

1. The Contractor shall be responsible for the design, shop drawings, manufacture, supply, installation, testing, commissioning, as-built drawings, maintenance manuals, training, warranties, and the maintenance of the Wall-mounted Davit Arm, including submission to the Architect for review without objection.
2. General Requirement
 - a. Davit
 - 1) Galvanized mild steel davit arms capable of rotating through 360°. Typical load ratings: 5.6 kN.
 - 2) Aluminum extruded sections of engineered length and size to suit application equipped with: carrying handle, wheel assembly, rolling trolley, friction trolley, upper turning arm, secondary hanger bracket, davit lock, prominently displayed non-corrosive data plates clearly stating Maximum Service Capacity of boom, Manufacturer's Name, and Installation Instructions. Designed to carry minimum vertical service load of 5.6 kN, i.e. rated load.
 - 3) Fabricate davits from heat treated structural aluminum alloy / GMS. Davit weight and configuration shall allow required movement to be accomplished by two people.
 - 4) Provide davits properly designed to mate with socket type, fixed, mobile, or flush-mount, indicated for proposed building location.
 - 5) All davits used for roof rigging shall be swung outside the building parapet to cover proposed maintenance area by two people standing on the platform.
 - 6) Provide davit rotation locks to prevent rotation in the davit bearing, either in the erect position or intermediate positions, while davit is in transition between horizontal and vertical, thereby preventing davit arm swing.
 - 7) Davits shall have adjustable rigging points, allowing variation in arm reach from the platform subsequent to rigging.
 - 8) Where davits with a reach in maximum out-reach distance are indicated on drawing, provide heavy duty type, capable of properly supporting oversize cantilever indicated.
 - 9) All pins and fittings subject to removal by workers shall be secured against loss by wire rope lanyards.
 - b. Davit Socket
 - 1) Davit Sockets: Fabricate from structural steel, galvanized after fabrication.
 - 2) The design and construction of the wall mounted davit socket is in accordance with code of practice for the structural use of steel 2005, Hong Kong to accommodate typical applied loads.
 - c. Powered Cradle
 - 1) Same type of powered cradle for System C - Monorail Track System with Powered Cradle can be used for this fixed davit arm system. Powered Cradle detail shall be refer to System C.

General Requirements	
Mounting Location	Wall-mounted Davit Arm at Flat Roof of Staircase S03
Permanent	Yes
Quantity	Refer to tender drawings
Warranty Period	10 Year

Design Life (Fatigue etc.)	10 Years
Defects Liability Period	12 Months
BMU	
Operation	with 2.4m self-climbing type cradle attached
Height from top of roof	Refer to Contract Drawing
BMU Type	Wall-mounted Davit Arm
Jibhead Type	Swivelling
Lifting Height	Refer to Contract drawing
Material	Galvanized mild steel painted to architect's requirement
Suspended Load	250kg S.W.L. for gondola cradle
Total Jib Reach	Refer to Contract Drawing
Height of Suspension Point	1.8m from A.F.F.L of flat roof of staircase S03
Outreach	1.4m nominal
Davit	
Gauge	Proposed by contractor
Davit Type	Single Davit Arm with Spreader
Material	Galvanized mild steel
Length	Refer to Contract drawing
Davit mounted Type	Parapet mounted
Power Supply	
Supply	380V 3 Phase, 32A, 50Hz
Sockets	PCE 5 pin 32 Amp

D. System C: Monorail Track System with Powered Cradle

1. The Contractor shall be responsible for the design, shop drawings, manufacture, supply, installation, testing, commissioning, as built drawings, maintenance manuals, training, warranties, and the maintenance of the Powered Cradle with monorail track, including submission to the Architect for review without objection.
2. The monorail system shall consist of monorail track mounted on brackets to the structure, out of the façade. Manual drive suspension trolleys shall be used to travel along monorail track, to which the powered cradle is to be rigged.
3. The monorail system and all necessary related work shall be designed and engineered in accordance with good engineering practice, EN1808 and with relevant Standards in order to provide a complete installation.
4. General Requirement
 - a. Monorail Tracks
 - 1) Track shall be manufactured from marine grade structural aluminium and anodized. Track systems shall be supplied by manufacturers with proven project references.

Tracks shall be designed for travelling loading including all the effects in accordance with EN1808. The contact points for the suspension trolleys shall be checked for local bending of section

E. System C: Monorail Track System with Powered Cradle

1. The Contractor shall be responsible for the design, shop drawings, manufacture, supply, installation, testing, commissioning, as built drawings, maintenance manuals, training, warranties, and the maintenance of the Powered Cradle with monorail track, including submission to the Architect for review without objection.
2. The monorail system shall consist of monorail track mounted on brackets to the structure, out of the façade. Manual drive suspension trolleys shall be used to travel along monorail track, to which the powered cradle is to be rigged.
3. The monorail system and all necessary related work shall be designed and engineered in accordance with good engineering practice, EN1808 and with relevant Standards in order to provide a complete installation.
4. General Requirement
 - a. Monorail Tracks
 - 1) Track shall be manufactured from marine grade structural aluminium and anodized. Track systems shall be supplied by manufacturers with proven project references.
 - 2) Tracks shall be designed for travelling loading including all the effects in accordance with EN1808. The contact points for the suspension trolleys shall be checked for local bending of section walls. Curved sections of track shall be checked for torsion due to the offset of the loading points and track support points.
 - 3) The track support brackets shall be checked for the worst case combined forces and moments applied to the tracks. Eccentric fixing conditions shall also be considered in the calculations.
 - 4) The track shall be mounted on brackets to the structure. The track shall be recessed into the soffit for System C with minimum visible opening.
 - b. Track Trolleys
 - 1) A number of suspension trolleys shall be provided for each monorail system to suspend the powered cradles.

- 2) An additional trolley shall be provided for suspension of a temporary glazing/material replacement hoist.
 - 3) Trolleys for System C shall remain permanently on the track.
 - 4) Movement along the monorail track shall be rope pull manual.
 - 5) The trolleys shall have manual braking system that can be engaged when in the working location.
 - 6) Each trolley shall have a working load capacity of 500 kg plus suitable safety factor in accordance with EN1808.
- c. Track Support Brackets
- 1) The monorail support brackets shall be fixed to structural support elements. The Contractor shall be responsible for the design, manufacture and installation of the support brackets for the monorail track. The Contractor shall be responsible for connecting the monorail track to the brackets, and the brackets to the structural support elements, including all fixings and anchors.
 - 2) Support brackets shall be hot dip galvanized.
 - 3) Fixings and anchors shall be fabricated from Grade 316 stainless steel and shall be supplied and installed by the Contractor. Any anchors requiring embedding shall be stainless steel and shall be designed and supplied by the Contractor for installation.
 - 4) The Contractor shall be responsible for coordinating for the positioning of the support brackets and shall provide all data necessary to evaluate the structural requirements for the structural elements to which the support brackets are to be installed.
 - 5) The location of the support brackets shall be defined Contractor. All support brackets shall be fixed to the structure at intervals justified by load calculations for the monorail system.
- d. Track Fixing
- 1) All bolted fixings shall be Grade 316 stainless steel. Measures shall be taken to prevent bimetallic corrosion.
 - 2) Tracks shall be attached to the brackets or structure with appropriate fixings of proper size and embedment (if required), which safely and adequately support the intended loads and transfer the loads to the structural support provided.
 - 3) Installation shall be in accordance with shop drawings certified by the manufacturer and the Registered Professional Engineer
 - 4) Reinforcement of structural members shall not be cut or damaged.
 - 5) Track Joint locations shall be within 500mm of support brackets and in accordance with manufacturer requirements. Joints shall be spliced in accordance with the track manufacturer's certified design.
 - 6) The Contractor shall provide all data necessary to evaluate the structural requirements for the structural elements to which the systems are to be installed.
- e. Powered Cradle
- 1) Each cradle shall consist of the cradle frame, two traction hoists and suspension ropes. The cradle shall be designed to carry 2 people together with tools and cleaning materials. Safe Working Load (SWL) shall be 240kg. Design and manufacture of the cradles shall comply with EN1808, local Authority requirements and this Specification.
- f. Operation of Powered Cradle

- 1) One primary powered cradle shall be provided for System C.
 - 2) The Contractor shall coordinate with the Architect as to the operational sequence.
- g. Cradle Suspension
- 1) Suspension ropes shall be connected to suspension points on the trolleys of the monorail systems when required.
 - 2) The cables shall be provided in one continuous length, be free from any crimp damage or other defects.
 - 3) The cradle shall be suspended by 4 numbers independent galvanized mild steel ropes.
 - 4) Hoist ropes shall be galvanized preformed steel. The suspension wire ropes shall, as a minimum, have a breaking strength of 2.5 metric tons and a design safety factor of 12 to 1.
 - 5) Suspension and safety ropes shall comply with all requirements of BS 6037/EN1808.
 - 6) The wire ropes shall not be less than 8mm in diameter.
- h. Cradle Rigging
- 1) The powered cradle shall be rigged to the monorail tracks within designated area on street level within site boundary or on accessible flat roofs as proposed by the Contractor.
- i. Sway Restraints
- 1) A sway restraint system shall be provided for the intermittent stabilization of the cradle's suspension ropes, complete with lanyards attached to restraint pins for attachment to the building restraint capsules.
 - 2) The restraint pins shall be stainless steel, quick release ball lock type device for insertion into the restraint capsule and shall be provided with the powered cradle.
 - 3) The lanyards shall be stainless steel cable, one-piece, with a means of adjustment for length.
 - 4) The restraint capsules shall be designed to suit the project requirements and shall be stainless steel with stainless steel fixings.
 - 5) The capsules and fixing bolts shall be fixed into the façade.
 - 6) The location of the sway restraint system shall be in accordance with the levels indicated on the Drawings and according to EN1808, subject to coordination to suit the façade design.
 - 7) The cradle restraint capsules shall be installed in order to protect the cradle from swaying due to wind, as per the following criteria:
 1. The lowest restraint point shall not be more than 40m above the lowest working level. The distance between restraints above 40m shall not exceed 15m.
 2. Lateral restraint capsules shall be located within the cladding and support structure and shall be designed to accommodate a 1.0kN load in any direction to accommodate wind generated sway.
 3. Restraint capsules located on the funnel shall be designed to accommodate a 3.0kN horizontal load to accommodate wind generated sway.

4. The restraint assembly shall be safely attachable and detachable by an operator on the cradle. The restraint assembly shall be designed so as not to cause any damage to the suspension wire ropes either when being fixed or removed when in operation.
- j. Haul-in Capsules on Funnel
- 1) Haul-in capsules shall be designed, manufactured, supplied and installed by the Contractor.
 - 2) The haul-in capsules shall be designed to suit the project requirements and shall be stainless steel with stainless steel fixings. The capsules and fixing bolts shall be fixed into the façade by the Contractor.
 - 3) The haul-in capsules are required to haul-in the cradle on the inwardly sloping recessed areas of the funnel façade. The locations shall be subject to final coordination with the project design. The levels indicated on the Drawings are indicative only and need to be confirmed and agreed by the Contractor in the final design.
 - 4) Tie-in lanyards shall be provided for intermittent hauling in of the cradle suspension ropes, complete with stainless steel pins for attachment to the building haul-in capsules. These shall accommodate the loads imposed by the haul-in of the cradle.
 - 5) Lanyards shall be stainless steel cable, one-piece, with a means of adjustment for length (hauling-in). The restraint pins shall be stainless steel, quick release ball lock type device for insertion into the haul-in capsule.
 - 6) The haul-in assembly shall be safely attachable and detachable by an operator on the System C cradle. Restraint/haul-in capsules located on the funnel shall be designed to accommodate a 3.0kN horizontal load to accommodate the cradle haul-in loads generated from the powered cradle as a result of the inward slope of the funnel.
 - 7) The system shall be designed so as not to cause any damage to the suspension wire ropes either when being fixed or removed when in operation.
- k. Power Supply
- 1) Power shall be provided for the Powered Cradle operation, as 3 phase 380V electrical power supply. Locations of the power points have been indicated on the Drawings.
 - 2) The final location of the power supply outlet for all systems shall be defined by the Contractor. The power supply for the system is intended to be provided along internal side of roof parapets which the system serves (refer to drawings) at minimum 20m interval.
 - 3) The final power supply outlets shall be located at positions confirmed by the Contractor. The power supply and waterproof outlets shall be installed as part of the Contract. The power supply system between the power outlet and the access system shall be designed, manufactured, delivered, installed and tested by the Contractor to ensure full operational integrity, including the co-ordination, design, supply and installation of all fixings necessary.
- l. Technical Features of Powered Cradle
- 1) The powered cradle shall be provided with rubber bumper strips and rubber rollers to protect the building façade. Powered Cradles shall be manufactured from the following materials
 1. Cradle structure and deck – Aluminum Anodized 25 microns
 2. Flooring – non-slip flooring complying with PTS Part B01 Clause 30-05-30/280 slip resistance

3. End frames – steel hot dip galvanized 85 microns

Required features:

1. Twin hoist motors with no power descent system
2. Over-speed safety device
3. Slack rope safety device
4. Electromechanical overload
5. Top and ultimate top limit switch
6. Safety belt attachment points
7. Cable bin
8. Twin drum wire winders
9. Soft wall bumper-rollers
10. Caster wheels
11. Bottom trip bar

Required operating controls:

1. Hold to run push buttons (up/down)
2. Hoist selector switch either side and both
3. Emergency stop button
4. Lockable main power switch
5. Bottom trip bar override button
6. Automatic self-leveling system
7. Power-on light
8. Resettable fuses
9. Thermal protector for wire winder motors
10. Hour meter
11. Phase protection – 3 phase only
12. Low voltage control circuit

Accessories to be included:

13. Original operation keys (2 sets)
14. 2 number safety harness (full body type) c/w lanyard
15. Operation and Service manual (3 Sets)

m. Particular Requirement for Monorail Track

General	
Monorail track system with powered cradle	As per Contract Drawings – Monorail tracks installation
Powered Cradle	One number of primary cradle
Permanent	Yes
Warranty Period	10 years
Design Life	25 years
Free Service Period	As per Defect Liability Period (DLP) – 12 Months
Monorail Tracks	
Mounting Location	As per Contract Drawings – 4 nos. of tracks at various levels

Location	Track located on brackets out of façade and coordinated with façade design
Rigging of Cradle	Track mounted on façade with cradle to be hooked to track at designated flat area on street level or flat roof within site boundary
Track Supplier	Proprietary system
Support Centres	1.5m (nominal), to suit Contractor proposal and to suit support brackets and façade module installation and all other building services and fixings to be installed on facade
Anchors and Fixings	Stainless steel (GMS not permitted). Bi-metallic protection required.
Support Brackets	Hot Dip Galvanized mild steel
Coordination	Required for location of brackets, support and restraint sockets
Loads	To suit Contractor's design per trolley
Installation	Only by manufacturer's authorized, trained personnel
Track Material	Marine Grade Aluminium, anodized
Type of Finish	Anodized, Fasteners and anchors to be stainless steel (Type 316)
Trolleys	5 number: 2 no. for each powered cradle plus one for materials delivery
Trolley Drive	Manual rope drive
Brakes	Manual brake on each trolley

n. Particular Requirements for Powered Cradles

General	
Mounting Location	On monorail tracks system as per Contract Drawings
Quantity of Cradle	One number of primary cradle
Permanent	Rigged with required
Warranty Period	10 years
Design Life	25 years
Free Service Period	As per Defect Liability Period (DLP) – 12 Months
Powered Cradle Requirements	
Type	Two man - Galvanized mild steel frame painted, and anodized aluminum body, painted
Length	Refer to Contract Drawings
Width	Refer to Contract Drawings
Controls	Cradle control panel for cradle vertical movement
Safe Working Load	240kg
Bottom Trip Bar	Yes, with bumper cover
Side Impact Bumpers	Required full length
End Impact Bumpers	Required full length
Lanyards	Required
Storage Location	To be confirmed.
Powered Cradle Hoist Requirements	
Hoist Type	Traction hoist
Number of Hoists	Two
Lifting Height	Refer to Contract Drawings
Wire Rope Laying	Multi

Wire Ropes	4 Independent ropes x 8mm min. dia. galvanised
Wire Rope Drums	Required
Anchor Points	Required in cradle
Number of Users	Two
Wire Rope Safety Factor	12
Controls	Cradle only
Hoist Speed	10m/min
Power Supply	380V 3 Phase, 32A, 50Hz
Power Source	Power cable connected to power outlets on roof parapet walls at max. 20m intervals
Sway Restraint and Pull-in Capsule Requirements	
Sway Restraints	Required as shown on Contract Drawing
Loading on Sway Restraints	1.0N in any direction
Haul-in Restraints	Required as shown on Contract Drawings
Loading on Haul-in Capsules	3.0kN outward 1.0kN in other directions
Coordination	Required with the design of the facade, panel openings, façade module installation and all other building services and fixings to be installed on facade
Capsule Type	To suit the design of the quick release pin, capable of accommodating design loads
Capsule Material	Stainless steel 316
Pin	Positive engaged bayonet ball-lock type pin, enabled for quick release
Lanyards	Adjustable, suitable for haul-in loads and ease of haul-in

F. Safety Devices

1. The BMU system shall incorporate a system of safety devices to ensure the safe operation of the system and the safety of the workers using the BMU system. All systems shall be designed to fail to safe mode.
2. The safety devices/system proposed shall comply with and shall be installed in accordance with all relevant Hong Kong by-laws, regulations and Ordinances current at the time of the date of return of this tender.
3. The Contractor shall be responsible for testing and commissioning of the safety devices and for obtaining all relevant certificates, reports or licenses from all relevant authorities that have jurisdiction over the installation and operation of the subject devices. All processing and testing fees and the like necessary to obtain the certificates, reports or licenses shall be borne by the BMU Sub-Contractor.

2.03 ANCILLARY MATERIALS

(NOT USED)

2.04 FABRICATION

- A. Fabrication shall be executed by a manufacturer specializing in the design and fabrication of Building Maintenance Equipment systems having a minimum of five years documented experience.

2.05 FINISHES/FINISHING TREATMENTS

(NOT USED)

2.06 STORAGE

(NOT USED)

2.07 WORKMANSHIP

(NOT USED)

PART 3 EXECUTION

3.01 EXAMINATION / INSPECTION

(NOT USED)

3.02 PREPARATION

A. Facilities on Site

The Contractor is to provide the following services free of charge and to afford all reasonable facilities for the proper execution of the work described in this contract.

The Contractor shall coordinate all trades and builder works required to be carried out as early as possible, and not later than three weeks prior to relevant sections of work commencing so that all chases, holes, openings, etc. may be coordinated, cut or formed as the work proceeds. All anchors and cast-in anchors shall be supplied by the Contractor.

The Contractor shall provide supervision on site to ensure that the setting out and position of anchors, holes, chases, and etc. are carried out to his requirements.

B. Coordination Between Trades

The contractor shall liaise between all sub-contractors and tradesmen who will be carrying out other services on the site, prior to the commencement of actual installation, so as to ensure the work of all trades are will be coordinated. No claims shall be entertained for non-compliances with the clause

3.03 INSTALLATION / APPLICATION

A. Good workmanship, proper material and proper site coordination is essential

B. All workmanship shall be subjected to the acceptance by the Architect who reserves the right to disapprove any part of the installation not complying with requirements stipulated in this specification or elsewhere under contract. The Contractor shall diligently carry out all necessary remedial work or replacement, free of charge and without any delay to the contract, on any item of work commented by the Architect.

C. No work shall be covered up or put out of view without the prior approval of the Architect. The Contractor shall provide minimum 7 working days' notice to the Architect for the examination and measurement of any work required to be covered up.

D. Endorsement by the Architect shall not relieve the Contractor of his responsibilities for the quality of works and materials as required under this contract.

E. All works to be completed by skilled and experienced workmen.

F. Properly coordinate with services of the various trades, carefully coordinate levels, sequence of works, and accessibility of various services.

G. Corrosion Protection

1. Components shall be constructed of heat-treated aluminium alloy, stainless steel, or hot-dip galvanized structural steel. Dissimilar metals, when used, shall be protected against electrolytic action. All connectors shall be Type 316 stainless steel, or stainless steel with equivalent or greater corrosion protection and strength, unless otherwise noted. Certified welders shall perform all welding work and all welds shall be examined by non-destructive testing (relevant certificates shall be provided to the Architect).

2. Components in contact with façades and the cradle casters shall be non-marking and scuff resistant material.
3. All carbon steel assemblies and components shall be hot-dip galvanized, 600 g/m² coating, in accordance with ASTM A123 and ASTM A153. These shall be galvanized after fabrication. Steel surfaces where burned by field welding or where damaged shall be cleaned and touched-up with 3 coats of zinc rich primer. In addition, steel shall be finished where exposed to view as specified herein.
4. Exterior finish of all roof cars and equipment and all other factory painted assemblies shall be weather, UV and salt resistant, exterior quality machinery enamel finished with colours as directed by the Architect.
5. Aluminium shall be anodized with minimum 18 µm anodic coating. All exposed aluminium assemblies shall be fluoropolymer finished to colours as directed by the Architect.
6. Stainless steel assemblies shall be of natural colour; mill finish.
7. All hardware, clips, bolts, nuts, washers, etc. shall be stainless steel, Type 316.

H. Painting

1. Painting of the Building Maintenance Equipment, if required shall be completed at the place of manufacture. The Building Maintenance Equipment shall be provided with a zinc flame sprayed anti-corrosive protection where appropriate.
2. Painting shall be to the colour defined by the Architect.
3. All paint colours shall be submitted to the Architect for review without objection.
4. Touch up painting shall be undertaken on site prior to handover using the same paint from the original supplier.

I. Electric Cabling / Wiring

1. Travelling cables for the Building Maintenance Equipment where required shall be of weatherproof type, with black heavy duty HOFR (Heat Resistant, Oil Resistant and Flame Retardant) sheathing complying with BS EN 50525 and BS 6708. They shall be tough, abrasion resistant, unaffected by sea water and strong sunlight, and suitable for flexible trailing application. They shall be suitably suspended to relieve any undue strain in the individual conductors.
2. Travelling cables for the Building Maintenance Equipment shall be weatherproof type, suitably suspended to relieve undue strain in the individual conductor. The travelling cable shall be provided with a device to attach to the anchor provided by others at the power outlet.
3. This shall relieve strain on the cable and prevent this pressure from being transferred to the electrical connection at the socket.
4. All electrical wiring inside the roof car control panel shall be PVC insulated cables to BS 6231 and Hong Kong equivalent standards.
5. The cables shall be made of either single or stranded annealed copper, or solid stranded aluminium, and insulated by PVC.
6. All cables and wires shall be colour coded with proper circuit identification.
7. Exposed wiring shall be drawn in G.I. conduit or flexible conduit with earth wire.

J. Electric Motors

1. Motors shall be totally enclosed fan cooled (TEFC) squirrel cage induction type and shall comply with BS EN 60034-1: 2006.
2. Motors shall be continuously rated to BS EN 60034-1: 2006.

3. Motors shall be insulated to BS EN 60084:2004 with class F as a minimum insulation.
4. All motors shall be protected with enclosures to at least IP55 for external operation.
5. Motors shall be fitted with anti-condensation heaters of single-phase pattern and arranged so that the heaters shall switch "OFF" automatically when the motor is started, and switch "ON" automatically when the motor is stopped.
6. Motors shall be sized to handle a minimum of 25% above the continuous rating of the driven load.
7. Motors shall be statically and dynamically balanced and have a maximum speed of 1,500 rpm.
8. Motors shall meet Hong Kong IE2 rating requirements.
9. Motors shall be provided with cast iron terminal boxes suitable for the connection of heavy duty flexible conduits or armoured cables. They shall be mounted onto the same bed as the machines they drive.
10. An emergency stop button shall be mounted on the cradle control panel, on the auxiliary control and at the main power switch on the Building Maintenance Equipment.

K. Welding

1. All welding by oxy-acetylene shall be in accordance with BS 4872- 1:1982 and Hong Kong equivalent standards and electrical arc welding to BS EN 1011 and Hong Kong equivalent standards.
2. Welding shall be provided with plain ends and trimmed to fair face joints for welding.
3. Joints shall be made with butt welds on site and shall be well aligned.
4. Each joint shall be of sufficient strength to withstand the desired stresses, thermal expansion and the weight imposed.
5. All welds shall be free from slag, of even thickness, regular contour, and well fused with parent metal, annealed and hammered on completion and finished smooth.

L. Cast-in Fixings and Anchor Bolts

1. All anchor bolts to be fixed to concrete structure shall be coordinated and identified by the Contractor.
2. Bolt sizes shall be sufficient to stand the imposed stress and shall have shear strength with a safety factor. Embedded anchors shall extend into the base slab, and embed lengths shall be designed to assume that the only strength of base slab concrete has adequate strength.
3. All anchor bolts and cast in anchors shall be rust free grade 316 Stainless Steel and with anti-corrosive treatment before installation.
4. All cast in anchors shall be designed and installed by the Contractor.

3.04 TOLERANCES

(NOT USED)

3.05 CLEANING

(NOT USED)

3.06 PROTECTION

- A. During the installations, the Contractor shall provide protection to all the works which may be damaged during the works of this Contract.
- B. The Contractor shall adequately protect its equipment, materials and finished and unfinished work on site against unauthorized interface, inclement weather, and other causes of damage until such time as the finished work has been taken over the Employer.
- C. The Contractor shall at all times keep its installation clean and tidy and shall remove from site at its own expense all surplus materials and temporary works as soon as they are no longer required.
- D. Prior to final handover to the Employer the Contractor shall perform all testing, adjustments, and the obtaining of any required approvals from Relevant Authorities.
- E. At such time as may be directed by the Employer, the Contractor shall remove all temporary protections, make good any damage incurred under the works of this section at no additional cost, thoroughly clean all surfaces with suitable cleaning agents, make final adjustments and handover the entire Building Maintenance Equipment installation leaving all in new, clean, and working order to the satisfaction of the Architect.

3.07 MAINTENANCE AGREEMENT

- A. Defect Liability Period
 - 1. All maintenance works must be carried out by direct labor; sub-contracting of routine maintenance works to a third party shall not be allowed.
 - 2. The Contractor shall without additional charge (including freight) to the owner, renew any parts which prove faulty from workmanship or materials of the equipment supplied during or within the warranty or defects liability period.
 - 3. All components replaced within the maintenance and service period or the Defects Liability Period (DLP) shall be separately warranted for an additional twelve (12) months starting from the day of the accepted replacement.
 - 4. The new parts for urgent replacement shall be delivered to site promptly by all means including by freight.
 - 5. A thorough examination shall be conducted before warranty period or DLP expiration. All latest defects are to be recorded with the architect and the Contractor is to make good all defects within 30 days of the inspection.
 - 6. Service shall include regular examination of the installation by competent and trained employees of the Gondola supplier, and shall include all necessary adjustments, greasing, oiling, cleaning, supplies, and parts replacement to keep the BMU in perfect working condition.
 - 7. The contractor shall maintain with the client a logbook recording all maintenance visits, inspected items, findings, and repairs
 - 8. This maintenance service shall be jointly provided by the Main Contractor and any the BMU sub-contractor.
 - 9. The maintenance frequency during the 12 months DLP shall be clearly stated but should not be less than once every quarter.

- B. Schedule Maintenance Service

The seasonal routine maintenance shall be as the seasonal maintenance schedule and shall include at the minimum the following and include all other works as requested by the Architect:-

- 1. Visual inspection of equipment including anchorage, suspension wire ropes and adjust braking system whenever required;

2. Check every suspension wire ropes;
3. Check all sheaves, pins, and rollers;
4. Lubricate all bearings, hinges, wheels, etc.;
5. Check and tighten any loosen bolts and nuts;
6. Check rail, if any, and storage frames;
7. Check intercom performance;
8. Check electrical contacts are clean and replace as required;
9. Touch up paint on BMU system;
10. Any other work to maintain the BMU functioning in safe and sound working condition;
11. Issue inspection form and relevant certificates by competent persons as required by the Factories and Industrial Undertaking Ordinance. The person and the inspection form format are to be submitted for the architect's approval prior to any site works;
12. Submit detailed monthly report to the Building Management Office or the Client's representatives for retention. The report shall at the minimum include:-
 - a. Inspection dates
 - b. Inspection items
 - c. Defects found
 - d. Defects rectified in previous report
 - e. Items replaced

C. Test Certificates

The service and maintenance work shall satisfy the requirement of this Contract and shall include at a minimum the following:-

1. To be responsible for the systemic examination and adjustment of the equipment of accordance with the Maintenance Proposal submitted by the Contractor and approved by the Architect;
2. Proper recording of all maintenance works;
3. Submissions to relevant authorities and statutory bodies all required documents, and the acquisition of all necessary certificates and forms.

D. Maintenance Proposal After Defects Liability Period

A separate quotation for ongoing maintenance of the system for an additional period of 5 years after the contract stated defects liability period shall be provided when tendering. The quotation shall detail items included and excluded.

END OF SECTION